

Project, results and Git inspection

Fashion-MNIST ANN | Omer Adnan | 5 October 2026

Reproducible classification workflow

Fashion-MNIST provides 60,000 training and 10,000 test images (28 x 28). An 80/20 stratified training split yields 48,000 training and 12,000 validation examples. Pixels are normalized by 255. The model is Flatten -> Dense(ReLU) -> Dropout -> Dense(10, softmax), trained with Adam and sparse categorical cross-entropy.

prepare.py saves raw arrays; preprocess.py normalizes and splits; train.py saves model.h5 and history.csv; evaluate.py writes test metrics and a confusion matrix. Hyperparameters come from params.yaml. Git tracks source/configuration; DVC caches artifacts in a Google Drive remote.

Measured v1 versus v2

Version	Dense units	Test accuracy	Test loss
v1	128	88.37%	0.338573
v2	256	88.43%	0.340889
Final	256	88.43%	0.340889

A3 - log variants (full captured outputs supplied)

```
git log --oneline --graph --all
  Shows compact commit graph and all refs.
git log --stat -3
  Shows changed-file summaries for three commits.
git log -p -1
  Shows the latest commit's patch.
git log main..dev
  Shows commits reachable from dev but not main.
```

Original history includes separate script/config commits, the hotfix merge and rebase, move/remove events, v1/v2 tags, and the collaboration merge. The branch-range output is reconstructed in an isolated clone with main=v1 and dev=v2, because final main already contains dev.

A4 - differences and staging

```
git diff                # unstaged tracked-file edits
git diff --staged       # edits selected for next commit
git diff main..dev      # compares branch tips
git diff main...dev     # merge base versus dev
```

In the replay, an added preparation comment appears first as unstaged, then in the staged diff. The v1-to-v2 branch comparison changes dense_units from 128 to 256, metrics and lock data. Two-dot and three-dot diff agree when main is an ancestor of dev. These staging/branch demonstrations are reconstructed, not original screenshots.

Repository: <https://github.com/OmerAdnan1/MLOPS> Assignment location: assignment_3. Repository naming is a deviation from the requested fashion-ann-pipeline repository name.

Git exercises and remote evidence

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A5 - stash (reconstructed isolated demonstration)

```
git stash push -m "Pause preprocessing edit..."
git stash list
  stash@{0}: On dev: Pause preprocessing edit...
git switch main; git switch dev; git stash pop
```

An uncommitted preprocessing comment is saved, main is inspected, and the edit is restored on dev. The replay's final diff confirms restoration. The original project also contains the earlier stash-related documentation commit.

A6 - rebase

A hotfix starts from main, changes the README heading, and is merged into main. dev is rebased onto that updated main. Original reflog entries record the rebase picks and finish; reconstructed before/after graphs accompany this report. Rebase recreates commit IDs, so the earlier published dev required force-with-lease.

```
git merge --no-ff hotfix
git rebase main
Original graph: hotfix merge 7553a8c -> dev commits
```

A7 - reset (original refs + reconstructed behavior)

```
git reset --soft HEAD~1
  M   assignment_3/src/prepare.py   [staged in replay]
git reset --hard HEAD~1
  HEAD is now at ...; working tree clean
```

Soft reset moves the branch while retaining index and working files. Hard reset also resets index and working files. Original reset-soft-demo/reset-hard-demo tags retain the student's throwaway commits; the original reflog records both resets. Replays were performed only in a disposable clone.

A8 - tracked move and deletion

```
7e814c0 Move project notes into docs using git mv
  rename ../project-notes.txt => ../docs/project-notes.txt
74fcd9b Remove obsolete scratch file using git rm
  delete mode 100644 assignment_3/obsolete-scratch.txt
```

C - DVC and Google Drive

DVC was initialized with --subdir inside assignment_3. Initial raw/processed/model/report artifacts were tracked using .dvc pointers, committed, and pushed. They were then transferred to pipeline ownership to avoid duplicate output definitions. OAuth settings are local; authentication tokens are not committed.

```
uv run dvc push -r gdrive_storage
  Everything is up to date.
Remote: gdrive://1cWQ1aXsd-0v_v3R9FreQfBMd-2w10D0W
```

Drive stores hashed objects under files/md5, not the original filenames. Captured push/status output is supplied instead of an original Drive-browser screenshot. Reference: <https://doc.dvc.org/user-guide/data-management/remote-storage/google-drive>

Pipeline configuration and reproduction

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Final dvc.yaml

```
stages:
  prepare:
    cmd: uv run python src/prepare.py
    deps:
      - src/prepare.py
      - pyproject.toml
      - uv.lock
    outs:
      - data/raw

  preprocess:
    cmd: uv run python src/preprocess.py
    deps:
      - src/preprocess.py
      - data/raw
      - pyproject.toml
      - uv.lock
    params:
      - preprocess.test_size
      - preprocess.seed
    outs:
      - data/processed

  train:
    cmd: uv run python src/train.py
    deps:
      - src/train.py
      - data/processed
      - pyproject.toml
      - uv.lock
    params:
      - train.dense_units
      - train.dropout_rate
      - train.learning_rate
      - train.epochs
      - train.batch_size
      - train.seed
    outs:
      - models

  evaluate:
    cmd: uv run python src/evaluate.py
    deps:
      - src/evaluate.py
      - models/model.h5
      - data/processed/x_test.npy
      - data/processed/y_test.npy
      - pyproject.toml
      - uv.lock
    outs:
      - reports
    metrics:
      - metrics.json:
          cache: false
```

Final params.yaml

```
preprocess:
  test_size: 0.2
  seed: 42

train:
  dense_units: 256
  dropout_rate: 0.2
  learning_rate: 0.001
  epochs: 20
  batch_size: 128
  seed: 42
```

D3 - original saved log

Running stage 'prepare'
Running stage 'preprocess'
Running stage 'train'
Running stage 'evaluate'
Updating lock file

The original forced, no-run-cache reproduction executed all four stages and generated dvc.lock. The complete original console log accompanies the report.

D4 - original saved log

prepare: skipped
preprocess: skipped
train: ran
evaluate: ran

Only train.dense_units changed. Training depends on that key; evaluation depends on the resulting model. Neither preparation nor preprocessing depends on it, so both were skipped.

metrics.json uses cache:false for Git-tracked metrics. The identical reports/metrics.json and confusion matrix are DVC-cached and uploaded. uv.lock is a stage dependency so environment changes invalidate affected stages. Final full configuration files are also supplied as evidence.

Collaboration conflicts and final state

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E1-E3 - independent code and data changes

From the shared processed-pointer baseline, teammate-sim divided pixels by 256; main divided by 254 and clipped to [0,1]. Each regenerated processed data, updated its pointer and pushed. Merging teammate-sim produced genuine conflicts in the same source lines and processed-data hashes.

```
CONFLICT (content): assignment_3/src/preprocess.py
```

```
CONFLICT (content): assignment_3/data/processed.dvc
```

Captured source conflict (excerpt)

```
<<<<<<< HEAD
x_train = np.clip(x_train.astype('float32') / 254.0, 0.0, 1.0)
x_test = np.clip(x_test.astype('float32') / 254.0, 0.0, 1.0)
=====
x_train = x_train.astype('float32') / 256.0
x_test = x_test.astype('float32') / 256.0
>>>>>>> teammate-sim
```

Captured processed-data pointer conflict

```
outs:
<<<<<<< HEAD
- md5: 80e622291076427c5ecdc0d5b84afdee.dir
=====
- md5: 969320c4e29eb97046a9f51690979d58.dir
>>>>>>> teammate-sim
  size: 219590768
  nfiles: 6
  hash: md5
  path: processed
```

E4-E5 - resolution and reproduction

The authoritative resolution restores /255.0 for both image sets, regenerates processed arrays, updates the pointer and runs dvc checkout. The merge retains both branch histories. Standalone processed tracking is then removed and the full preprocessing stage restored from v2; dvc repro verifies the final four-stage workflow.

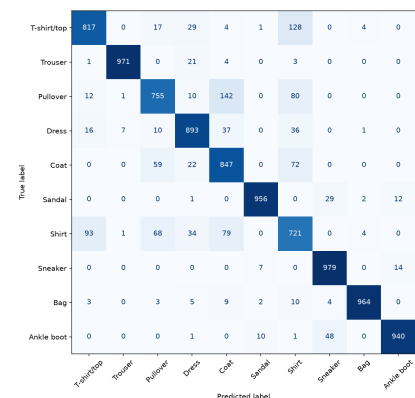
Final test accuracy: 88.43% (target >=85%)

Final test loss: 0.340889

DVC status: data and pipelines up to date

DVC push: final artifacts uploaded

Evidence note: original screenshots were not taken. Existing logs/history are preserved; replayed demonstrations are explicitly labelled. Conflict records and final-state checks are from actual assisted completion. Rendered output panels and full text files are included as supplementary evidence.



Submission: repository link, this four-page PDF, supplementary evidence and the Drive remote. Instructor access must be granted to their specific Google account; no instructor address was supplied, so sharing permissions were not changed.